

Calculus I Lesson 25

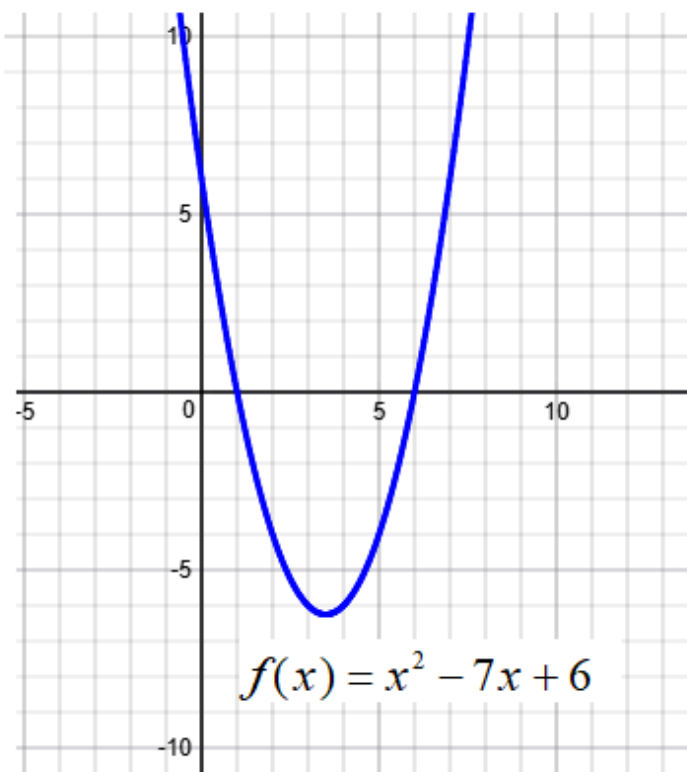
1) Find the minimum value of the function $f(x) = x^2 - 7x + 6$

a) $f(x) = x^2 - 7x + 6$ has a minimum at the point ____?

b) $f'(x) =$ ____?

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)



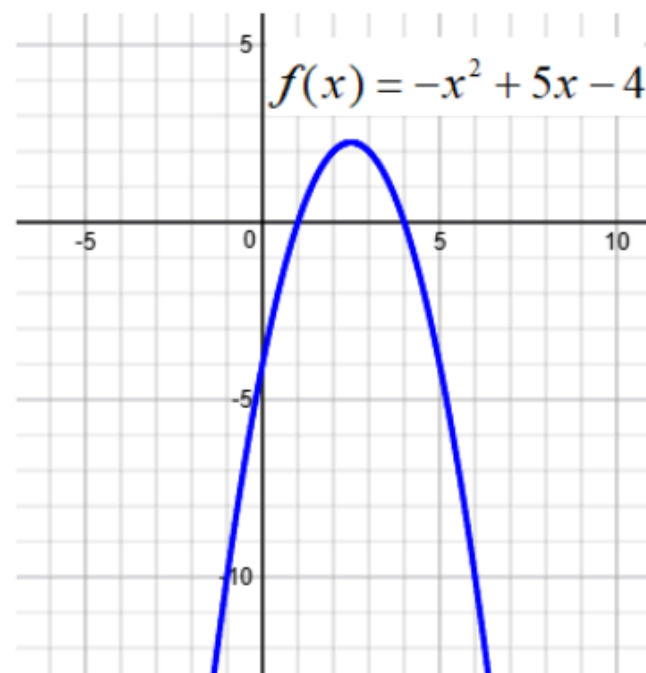
2) Find the maximum value of the function $f(x) = -x^2 + 5x - 4$

a) $f(x) = -x^2 + 5x - 4$ has a maximum at the point _____?

b) $f'(x) =$ _____?

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)



3) Find the minimum/maximum value of the function $f(x) = 0.7x^3 + 2x^2 - 5x - 4$

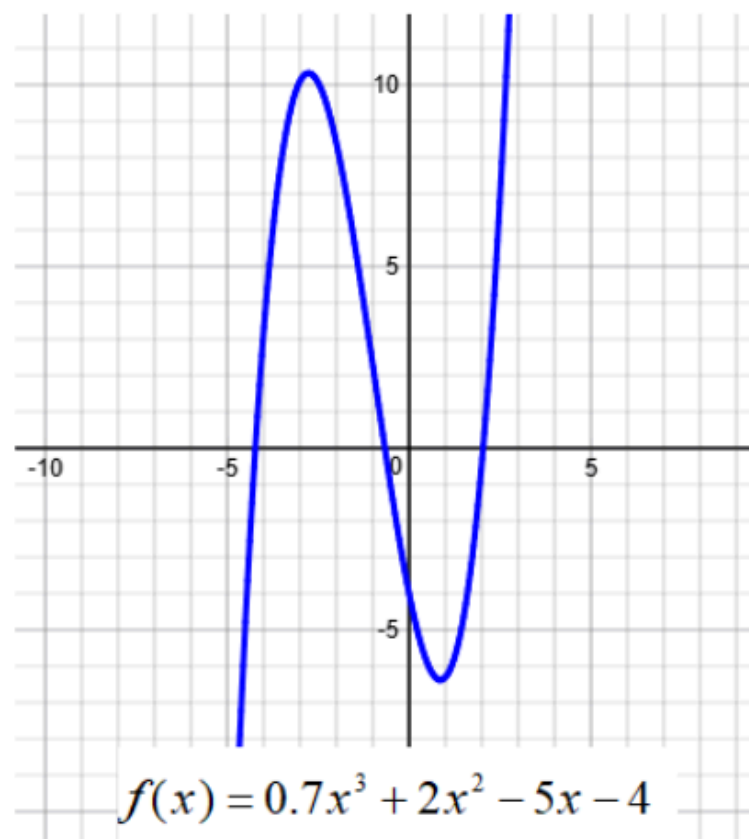
a) $f(x) = 0.7x^3 + 2x^2 - 5x - 4$ has a relative minimum at the point ____?

b) $f(x) = 0.7x^3 + 2x^2 - 5x - 4$ has a relative maximum at the point ____?

b) $f'(x) =$ ____?

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)



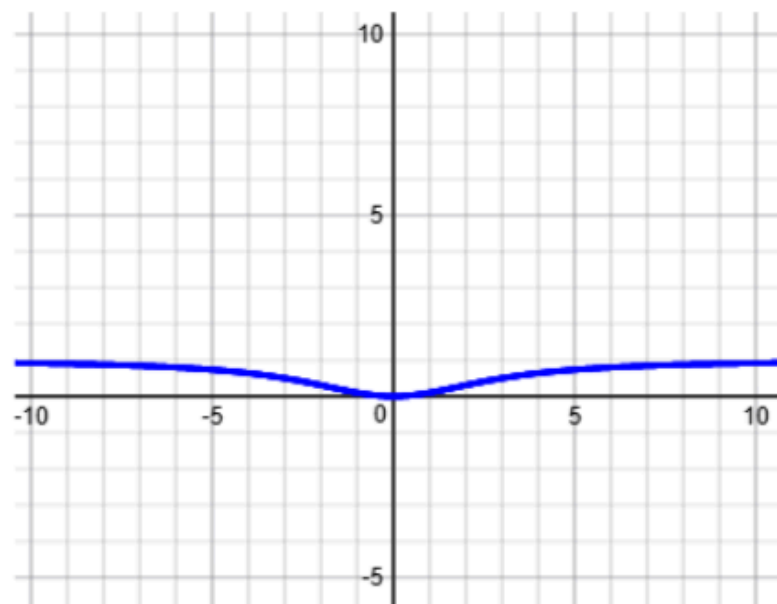
4) Find the minimum function $f(x) = \frac{x^2}{x^2 + 9}$

a) $f(x) = \frac{x^2}{x^2 + 9}$ has a minimum at the point _____

b) $f'(x) =$ _____

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)



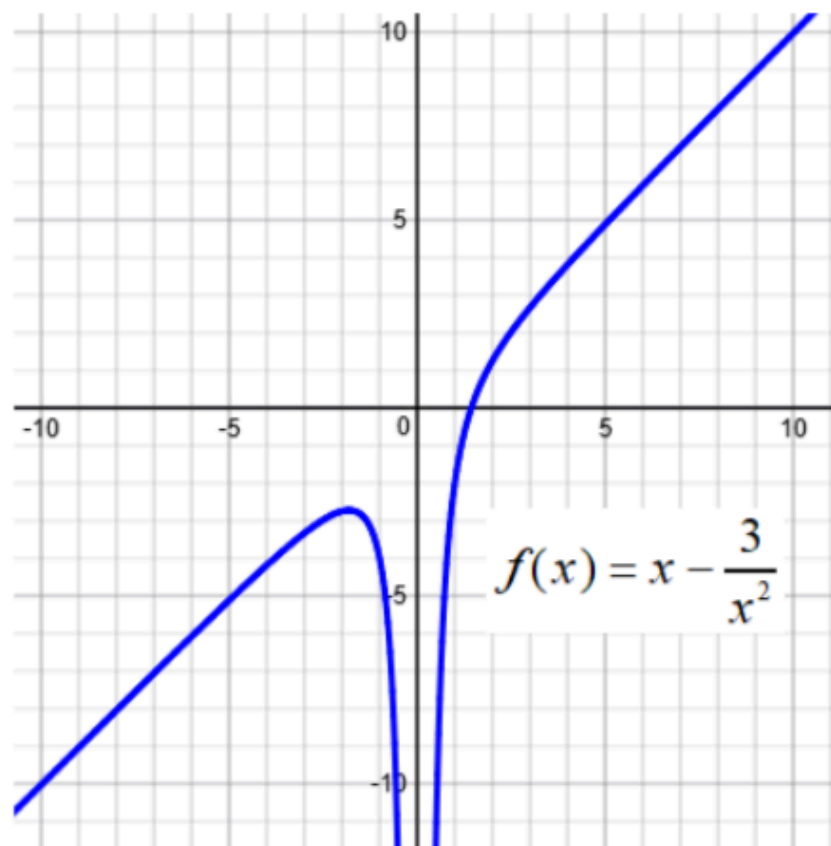
5) $f(x) = x - \frac{3}{x^2}$ has a relative maximum between $x = -2$ and $x = -1$.

a) Find the point where $f(x) = x - \frac{3}{x^2}$ has a relative maximum.

b) $f'(x) = \underline{\hspace{1cm}}?$

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)



6) Find the minimum function value of $f(x) = (x - 4)^{2/3}$

a) $f(x) = (x - 4)^{2/3}$ has a minimum at the point ?

b) $f'(x) = \underline{\hspace{1cm}} ?$

c) Set $f'(x) = 0$ and solve for x .

(Note: This is a critical number of $f(x)$)

